NUMERICAL METHODS FOR PDES
Lecture 5

Again, consider

$$\begin{cases} -u'' = f & \text{in } (0,1) \\ u(0) = 0 = u(1) \end{cases}$$

Weak form: $u \in V := H_0^1(0,1)$ such that

$$a(u,v) = \langle f, v \rangle \quad \forall v \in V$$

$$a(w,v) = \int_0^1 w' v' \, dx$$

Linear FEM:

- $V_h \subset V$ piecewise-linear functions w.r.t. partition $\{x_i\}_{i=0}^N$
- restrict weak form in V_h

$\Rightarrow$ algebric form

$$AU = F$$

$$A = \frac{1}{h}\begin{bmatrix} \ddots & \ddots & \ddots \\ & -1 & 2 & -1 \\ & & \ddots & \ddots & \ddots \end{bmatrix}$$

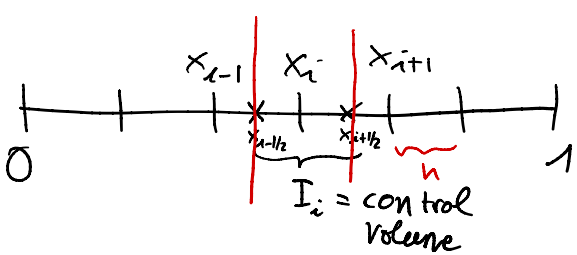
$$F_i = \int_0^1 f \varphi_i \approx h f(x_i)$$

φ_i: basis function

trapezoidal rule

Finite Volume

- again, consider partition (uniform)



1) Integrate PDE over I_i:

$$= -\int_{x_{i-1/2}}^{x_{i+1/2}} u'' \, dx = \int_{x_{i-1/2}}^{x_{i+1/2}} f \, dx$$

$$= -u'(x_{i+1/2}) + u'(x_{i-1/2})$$

flux

- balance ($\Rightarrow$ conservation)
- only first derivatives are involved!

2) approximate the balance law:

- u' by FD

$$u'(x_{i+1/2}) \approx \frac{u(x_{i+1}) - u(x_i)}{h}$$

$$u'(x_{i-1/2}) \approx \frac{u(x_i) - u(x_{i-1})}{h}$$

- $\int_{x_{i-1/2}}^{x_{i+1/2}} f(x)\,dx \approx h\, f(x_i) =: f_i$

(1) by rectangle rule

3) Fix grid function $\{U_i\}$

$$-\frac{U_{i+1} - U_i}{h} + \frac{U_i - U_{i-1}}{h} = h f_i$$

$$\boxed{\frac{-U_{i+1} + 2U_i - U_{i-1}}{h} = h f_i}$$

FD in several dimensions

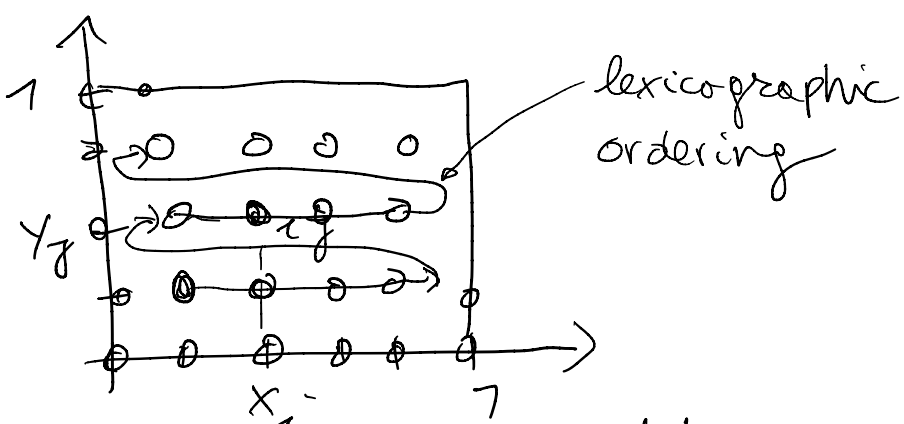
Model problem: in $\Omega = (0,1)^2$

Poisson: $\Delta u = u_{xx} + u_{yy}$

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

1) Fix $N = N_x = N_y$, (x_i, y_j)

$x_i = ih$; $y_j = jh$; $h = \frac{1}{N}$



2) approximate $u(x_i, y_j) \sim U_{ij}$

$$\Rightarrow \quad U = \{ U_{i,j} \}_{i,j=0}^{N}$$

3) $$u_{xx}(x,y) \approx \left(\delta_h^x\right)^2 u(x,y)$$

$$u_{yy}(x,y) \approx \left(\delta_h^y\right)^2 u(x,y)$$

4) (FD)
$$\begin{cases} U_{i,j} = 0 & (i,j) \text{ on boundary} \\ -\left(\delta_h^x\right)^2 U_{ij} - \left(\delta_h^y\right)^2 U_{ij} = F_{ij} & 1 \le i,j \le N-1 \end{cases}$$

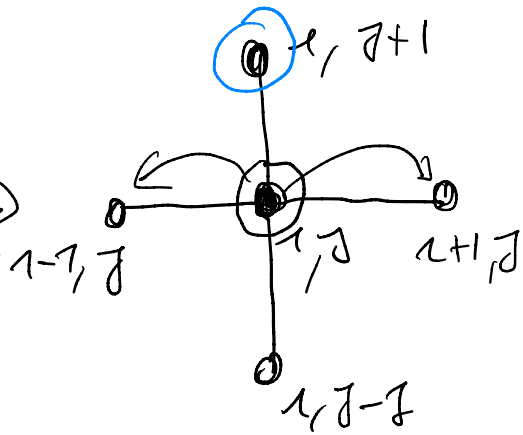
$\Delta_h U_{ij}$

where $F_{ij} = f(x_i, y_j)$.

$$\Delta_h U_{ij} = -\left[\frac{U_{i+1,j} - 2U_{ij} + U_{i-1,j}}{h^2} + \frac{U_{i,j+1} - 2U_{ij} + U_{i,j-1}}{h^2} \right]$$

$$= -\frac{U_{i+1,j} + U_{i,j+1} - 4U_{ij} + U_{i-1,j} + U_{i,j-1}}{h^2}$$

STENCIL of the FD operator →



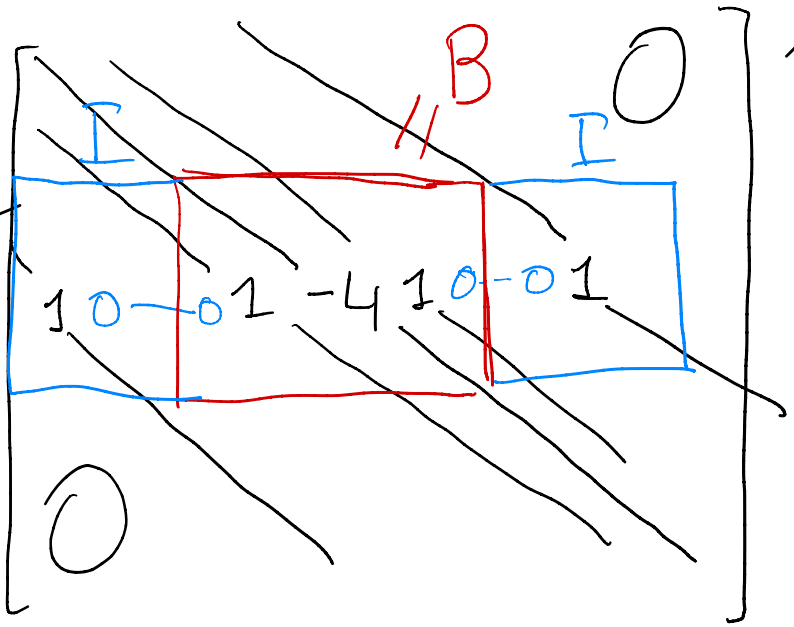
5) Algebric form:

order unknowns using lexicografic ordering

$$\underline{U} = \{ U_{1,1}, U_{2,1}, U_{3,1}, \dots, U_{M-1,1}, U_{1,2}, U_{2,2}, \dots, U_{M-1,2}, \dots, U_{1,M-1}, U_{2,M-1}, \dots, U_{M-1,M-1} \}$$

$$\underline{F} = \{ f_{1,1}, f_{2,1}, \dots \}$$

$$A = -\frac{1}{h^2}$$



penta diagonal matrix

(1,1)

$$= -\frac{1}{h^2} \begin{bmatrix} B & I & & & \\ I & B & I & & \\ & I & B & I & \\ & & \ddots & \ddots & \ddots \end{bmatrix}$$

$I = (M-1)^2$ identity ; $B = \begin{bmatrix} -4 & 1 & & 0 \\ 1 & \ddots & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 1 & -4 \end{bmatrix}$ $M-1$ $M-1$

Consistency

Truncation error:

$$T(x,y) = \mathcal{L}u(x,y) - \mathcal{L}_h u(x,y)$$

$$= f(x,y) - \mathcal{L}_h u(x,y)$$

$$= f(x,y) - \frac{1}{h^2}\Big[u(x+h,y) + u(x,y+h) - 4u(x,y) + u(x-h,y) + u(x,y-h)\Big]$$

Taylor (x,y)

$$= f(x,y) - \frac{1}{h^2}\Big[$$

$$u + h u_x + \frac{h^2}{2}u_{xx} + \frac{h^3}{6}u_{xxx} + \frac{h^4}{24}u_{xxxx}(\xi^+, y)$$

$$+ u + h u_y + \frac{h^2}{2}u_{yy} + \frac{h^3}{6}u_{yyy} + \frac{h^4}{24}u_{yyyy}(x, \xi^+)$$

$$- 4u$$

$$+u - h u_x + \frac{h^2}{2} u_{xx} - \frac{h^3}{6} u_{xxx} + \frac{h^4}{24} u_{xxxx}(\xi^-, y)$$

$$+u - h u_y + \frac{h^2}{2} u_{yy} - \frac{h^3}{6} u_{yyy} + \frac{h^4}{24} u_{yyyy}(x, \xi^-)$$

where $\begin{cases} u \to u(x,y) \\ \xi^+ \in (x, x+h);\ \xi^- (x-h, x) \\ \eta^+ \in (y, y+h);\ \eta^- (y-h, y) \end{cases}$

$$= f - \frac{1}{h^2}\left[h^2 u_{xx} + h^2 u_{yy} + \frac{h^4}{24}\left(u_{xxxx}(\xi^+, y) + u_{xxxx}(\xi^-, y) + u_{yyyy}(x, \eta^+) + u_{xxxx}(x, \eta^-) \right) \right]$$

PDE

$$= h^2 \left(u_{xxx}(\xi^+, y) + \dots \right)$$

Assume $u \in C^4([0,1])$

$$M_{xxxx} = \max_{(x,y)} u_{xxxx}(x,y)$$

$$M_{yyyy} = \max_{(x,y)} u_{yyyy}(x,y)$$

$$|T(x,y)| \leq \frac{1}{12} \boxed{h^2} \left(M_{xxxx} + M_{yyyy}\right)$$

$\Downarrow$ 2nd order